

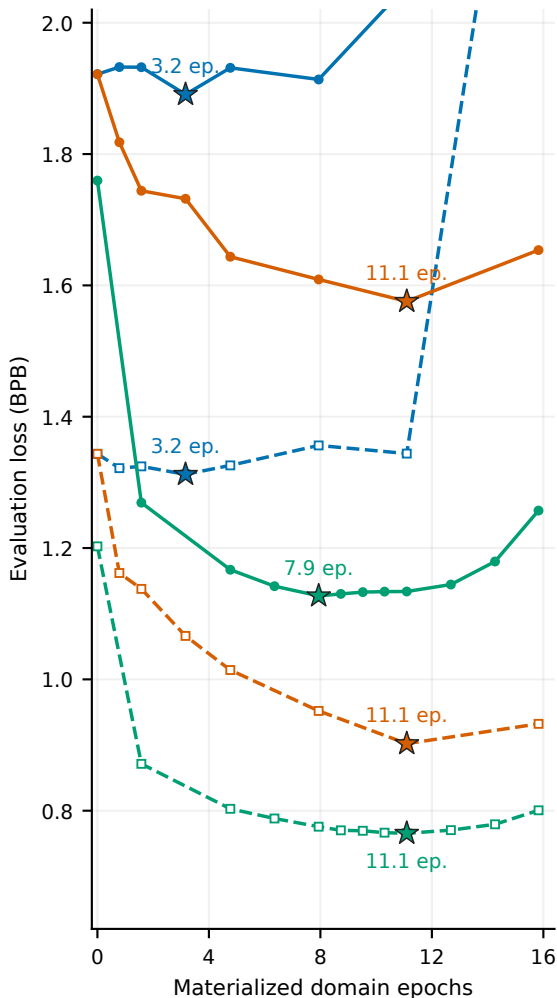
Wikipedia FineMath-3+ StarCoder

Simulated-epoching proxy (2.49×10^{16} FLOPs) Target (6.66×10^{18} FLOPs)

★ Measured minimum ♦ Fixed epoch count (1.6, 4.8, 7.9, 11.1) ♦ Simulated-epoching selection

Proxy and target sweeps

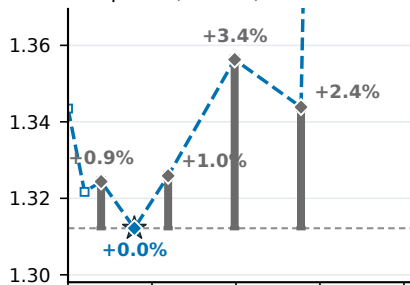
Wikipedia: GSM8K | FineMath-3+: GSM8K
StarCoder: Paloma code



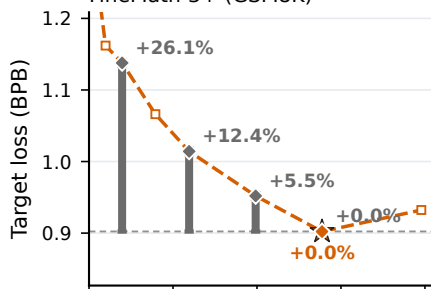
Fixed epoch count vs. simulated epoching

Excess target loss over the measured minimum

Wikipedia (GSM8K)



FineMath-3+ (GSM8K)



StarCoder (Paloma code)

